

IoT-Enabled Smart Soil Monitoring System with Crop and Fertilizer Recommendations

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Abstract— Soil health, influenced by variables such as temperature, moisture content, and nutrient levels, plays a crucial role in agricultural productivity. The proposed Smart Soil Monitoring System that integrates the Internet of Things (IoT) with machine learning to enable real-time soil analysis and provide actionable crop and fertilizer recommendations. The system utilizes various sensors, including NPK, soil moisture, temperature, and humidity sensors, interfaced with the Arduino UNO microcontroller. The MAX485 module ensures reliable data transmission, while an LCD module displays local soil parameter measurements. Data is sent to the cloud using the ESP8266 microcontroller, where it is continuously monitored, pre-processed, and analyzed. The Random Forest algorithm is employed to train a machine learning model using key soil variables identified through a feature extraction process. The FastAPI framework efficiently processes the data in the cloud, delivering real-time recommendations for optimal crops and fertilizers based on specific soil conditions. System performance is evaluated using metrics like accuracy and efficiency, demonstrating predictive reliability with 90% accuracy in crop prediction. By combining IoT and machine learning, the proposed system offers a scalable and cost-effective precision agriculture solution, providing farmers with data-driven insights to enhance agricultural yields, minimize resource wastage, and promote sustainable farming practices.

Keywords— IoT, Smart Agriculture, Soil Monitoring, Arduino UNO, ESP8266, Machine Learning, Crop Recommendation, Fertilizer Recommendation etc.

I. INTRODUCTION

With agrifood systems providing for the most of the people in the world, agriculture continues to be a vital industry for both economic growth and global food security. A key factor in agricultural productivity, soil health affects crop yield and sustainability. However, sustaining soil quality is severely hampered by land degradation, which is made worse by unsustainable farming methods and climate change, especially in places like Bangladesh. IoT-enabled soil monitoring systems, which offer real-time insights into soil parameters to drive precision agriculture, are one example of a new solution to these problems.

Difficulties with Earlier Works Even with significant progress in soil monitoring devices, a number of issues still exist. Scalability, dependability, and integration with predictive models are issues that many current systems, including those that use Wireless Sensor Networks (WSNs), must deal with. Additionally, farmers frequently make less-than-ideal decisions due to a lack of real-time data analysis and practical advice. Furthermore, the majority of systems do not include thorough evaluations of soil fertility, which restricts their use in a variety of agricultural contexts.

The Work's Motivation This research is strongly motivated by the urgent need for sustainable farming practices as well as the potential for IoT and machine learning (ML) to revolutionize soil monitoring systems. This effort attempts to close the gap between technology breakthroughs and real-world precision agriculture applications by tackling the difficulties of real-time data gathering, processing, and actionable insights. Scalable and effective solutions for enhancing soil health management and maximizing crop yield can be obtained through the integration of IoT, WSNs,

and ML.

This paper's main goals are to:

1. Develop and put into use an IoT-enabled smart soil parameter observation system that can collect and transmit data in real time.
2. Develop a machine learning model that employs soil factors in order to effectively suggest crops and fertilizer.
3. Evaluate whether the proposed solution meets system precision agriculture criteria when applied toward developing accuracy, efficiency, and user usability.

The following contributions are made by this paper:

1. Offers a complete Internet of Things architecture that integrates sensors for measuring NPK, temperature, humidity, and soil moisture and interfaces them with the Arduino UNO microcontroller and send data to cloud using ESP8266 Wi-Fi microcontroller.
2. Uses the MAX485 module to implement a dependable data transmission mechanism and offers real-time monitoring.
3. Uses information about soil properties to build and assess a machine learning model for crop and fertilizer recommendations.
4. Showcases the system's affordability and scalability for application in various agricultural contexts.

This paper's structure is as follows: A thorough analysis of relevant work is provided in Section 2, emphasizing any gaps in the body of knowledge. The architecture of the suggested system, including its hardware and software implementation, is covered in Section 3. System performance and experimental findings are covered in Section 4. The work is finally concluded in Section 5 with important findings and suggestions for further research.

II. RELATED WORKS

C. Emsden (2023) [1] This source highlights the vast global reliance on agrifood systems, highlighting how they provide for most of the people in the world. It emphasizes how important it is to utilize sustainable and effective farming methods in order to guarantee food security and financial stability for rural people. The information is useful for comprehending how soil monitoring systems might improve agrifood productivity in general.

Cavallito, M. (2023) [2] The serious effects of land degradation on food security in Bangladesh, a nation heavily

reliant on agriculture, are covered in this article. In line with the goals of intelligent soil monitoring systems, it highlights how crucial soil health management is to halting land deterioration and enhancing agricultural productivity.

D. J. Field and L. E. Pozza (2020) [3] investigate how food security and soil security are related. They contend that maintaining the quality of the soil is crucial to producing food in a sustainable manner. The scientific basis for comprehending how soil monitoring technologies contribute to soil and food security is provided in this study.

T. Arumuganathan and R. Madhumathi (2020) [4] A wireless sensor network (WSN) for measuring soil moisture and NPK is presented in this paper. It emphasizes how well WSNs work at delivering real-time soil data, which is essential for crop management and precision farming.

P. Akshay and U. K. Madhura (2017) [5] The application of WSNs for managing soil quality is covered in this research, with an emphasis on how they can lower manual labor costs and boost agricultural output. The suggested approach places a strong emphasis on data-driven decision-making and real-time monitoring.

S. Nugraha and L. Kamelia (2019) [6] A real-time soil fertility parameter monitoring system is suggested by this study. It provides a model for real-time soil analysis and crop suggestion, illustrating how IoT-enabled systems can enhance smart farming applications.

H. Pallevada et al. (2021) [7] The practical uses of IoT and machine learning in smart agriculture are demonstrated by this study, which focuses on real-time soil nutrient detection and analysis. It offers guidance on creating efficient soil monitoring systems.

M. Pyngkodi et al. (2022) [8] An IoT-based system for assessing soil nutrients is presented in the research, emphasizing how crucial it is to increasing agricultural output. It talks about combining sensors and Internet of Things platforms to accomplish effective soil monitoring.

Bolla, D.R. et al. (2019) [9] This study investigates the measurement of soil quality using image processing and the Internet of Things. It draws attention to how cutting-edge technologies can be used in conjunction with conventional soil analysis techniques.

Senthil, K.M. et al. (2023) [10] In order to integrate machine learning for effective agricultural operations,

suggest an Internet of Things-based smart farming solution. The creation of data-driven soil monitoring systems is encouraged by this study.

Mary, D.S. et al. (2022) [11] The use of deep learning and machine learning in smart farming is covered in this study. In line with the goals of IoT-enabled soil monitoring, it emphasizes their function in crop and fertilizer recommendations.

Ingle, A. (2023) [12] The training of machine learning algorithms to suggest crops depending on soil conditions is made possible by this dataset. It is essential for creating predictive algorithms for intelligent farming systems.

Rodriguez-Galiano, V.F. et al. (2012) [13] studied the efficacy of the Random Forest (RF) classifier in land-cover classification from remotely sensed data. The research sought to determine the strength of the RF algorithm in mapping intricate land cover and land use classes. It also sought to examine the algorithm's capability to process high-dimensional data and its performance relative to conventional classification techniques.

Hong, Z., Kalbarczyk, Z., and Iyer, R.K. (2016) [14] proposed a data-driven approach for soil moisture collection and prediction to enhance precision agriculture. The study focused on improving data acquisition efficiency and predictive modeling accuracy for soil moisture dynamics.

Ahmmmed et al. (2018) [15] The handbook is a thorough resource for fertilizer recommendations based on soil nutrient levels, which is in line with smart soil monitoring systems' objectives of increasing agricultural output.

III. PROPOSED METHOD

The suggested approach entails creating an IoT-enabled smart soil parameter observation system that combines a number of sensors to evaluate important soil characteristics like temperature, humidity, soil moisture, and NPK levels. An Arduino UNO microcontroller, which acts as the central processing unit is interfaced with these sensors and ESP8266 enables Wi-Fi-based data transfer. The system uses the MAX485 module, which permits long-distance data transfer across sensors, to provide dependable data connectivity. After being pre-processed, the gathered soil data is sent to an cloud platform for storage and real-time monitoring. Key soil properties are identified through a feature extraction method, and these are subsequently fed into a ML model which is already trained to select the best crops and fertilizers. Additionally, users on-site are presented with real-time soil information via a local display

unit that consists of an LCD. With its emphasis on scalability, cost-effectiveness, and user-friendliness, the suggested approach is appropriate for a variety of agricultural applications.



Fig. 1. Architecture of the proposed method

A. Hardware Description

1) *ESP8266 Microcontroller*: It is a microcontroller with integrated Bluetooth and Wi-Fi. As the CPU, it manages sensor data collection and sends it to the IOT platform. It works at the voltage of 2.5 volts.



Fig. 2. ESP8266 Microcontroller

2) *Humidity Sensor (DHT11)* : In order to evaluate the environmental factors influencing soil health, this sensor detects atmosphere humidity. It is essential for figuring out the best times for irrigation and agricultural growth. It works at the voltage of 3 to 5.5 volts.



Fig. 3. Humidity Sensor

3) *Soil Moisture Sensor (REES52)*: Real-time information on soil hydration levels is provided by the soil moisture sensor, which measures the soil water content. By using this data, irrigation techniques can be optimized to avoid overwatering or underwatering. It operates at the voltage of 5 volts.

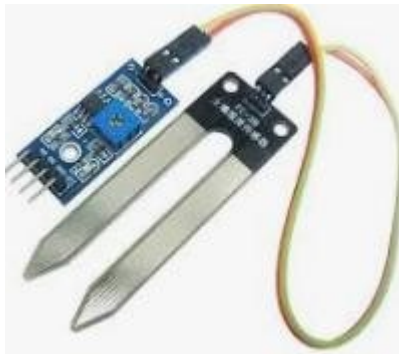


Fig. 4. Moisture Sensor for Soil

4) *NPK Sensor (LNPK-101)*: NPK sensor gauges the soil's concentrations of potassium (K), phosphorus (P), and nitrogen (N). Plant growth depends on these minerals, and their real-time monitoring guarantees accurate fertilizer delivery, cutting waste and boosting soil fertility. It operates at voltage 12 to 24 volts.



Fig. 5. NPK Sensor

5) *Soil Temperature sensor (DS18B20)*: It is used to find the temperature of the soil, allows farmers to optimize planting times. It operates at 5 volts.



Fig. 6. Soil Temperature sensor

6) *MAX485 Module*: For dependable, long-distance data transfer between sensors and the microcontroller, the MAX485 is a low-power RS-485 transceiver. It guarantees less data loss and effective transmission.



Fig. 7. MAX485 Module

7) *LCD Display (RG1602)*: A user-friendly interface for on-site real-time soil parameter visualization is offered by the LCD display. Farmers don't need external equipment or internet connectivity to get vital data. It operates at voltage of 5 volts.



Fig. 8. LCD Display

8) *Arduino UNO*: Arduino UNO is a microcontroller board that reads sensor data via its analog and digital pins; it processes the read data

and sends out the data to other devices. It commonly finds application for real time monitoring in IoT and automation projects.



Fig. 9. Arduino UNO

B. Software Description

The suggested system's software is essential for communication, data processing, and analysis. It consists of three primary modules: sensor interfacing, IoT integration, and machine learning-based recommendation creation. The sensor interfacing module is managed using an Arduino Uno, with the ESP8266 acting as a Wi-Fi module for wireless communication. Sensor data is gathered, pre-processed, calibrated, and formatted using the Arduino IDE to ensure accuracy and consistency. The IoT integration module utilizes the ESP8266's Wi-Fi functionality to transmit real-time data to a cloud-based platform. This platform provides a user-friendly dashboard for remote monitoring and data visualization. The machine learning module uses the Random Forest algorithm to analyze soil data and generate crop and fertilizer recommendations. A large dataset is used to train the model, ensuring high accuracy and reliability. The FastAPI framework is used to create API for seamless integration and communication between the system components. Together, these modules enable efficient system operation and provide valuable insights to enhance agricultural productivity.

Random Forest Algorithm:

The IoT-enabled smart soil monitoring system collects real-time soil parameters, including Nitrogen (N), Phosphorus (P), Potassium (K), temperature, and humidity, using integrated sensors. This data is transmitted to the cloud, where it undergoes preprocessing before being fed into a trained Random Forest model. The model, built using a large dataset of historical soil properties and corresponding crop yields, generates multiple decision trees to analyze patterns and relationships between soil characteristics and optimal crop selection. By employing a majority voting mechanism, the Random Forest model predicts the most suitable crop for the given soil conditions with an accuracy exceeding 90%. This approach ensures data-driven decision-making, enabling farmers to maximize crop yield, optimize

soil health management, and support sustainable farming practices.

C. Work Flow

1. System Initialization

- Power on the system and initialize the Arduino UNO and ESP8266 microcontroller.
- Set up Wi-Fi connectivity for IoT communication.

2. Sensor Configuration

- Calibrate sensors for soil parameters.
- Establish communication with the sensors using appropriate protocols.

3. Data Collection

- Collect real-time data from all connected sensors.
- Store the raw data in a temporary buffer.

4. Data Pre-processing

- Filter out noise and outliers from the collected data.
- Normalize sensor readings to ensure consistency.

5. Data Transmission

- Transmit pre-processed data from the ESP8266 to the cloud platform using Wi-Fi.
- Log data in a cloud-based database for analysis and monitoring.

6. Feature Extraction

- Analyze the collected data to extract key features such as average soil moisture and nutrient levels.

7. Machine Learning Model Prediction

- Input the extracted features into the pre-trained machine learning model.
- Generate crop and fertilizer recommendations based on soil conditions.

8. Data Visualization

- Display real-time soil parameters on the LCD.
- Provide a graphical representation of data on the web app dashboard.

9. User Notification

- Send recommendations to the farmer through the cloud platform.

10. System Optimization

- Continuously monitor the system's performance.
- Update the machine learning model periodically with new data and user feedback.

D. Implementation Flow chart

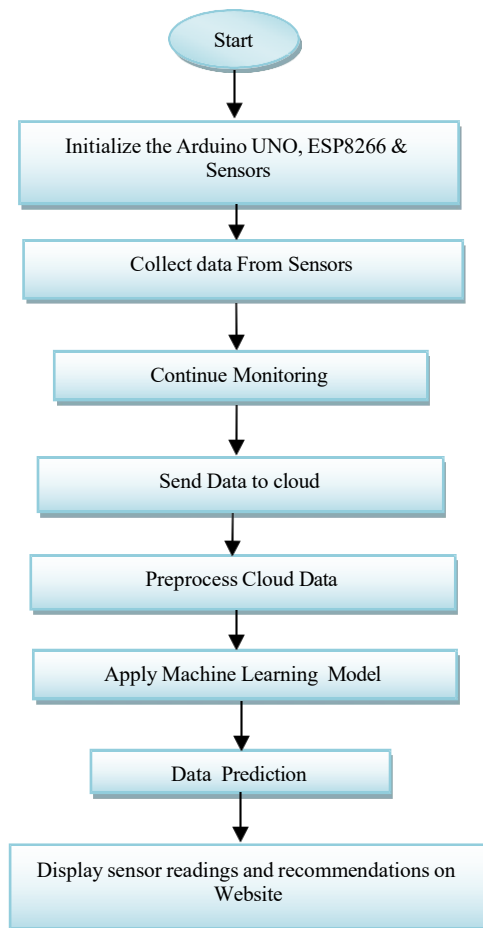


Fig. 10. Implementation flow chart

IV. EXPERIMENTAL RESULTS

A. System Performance Evaluation

The outcomes of the experiment show how successful the suggested IoT-enabled smart soil monitoring system is in actual farming environments. The system's sensors were used to measure important soil factors. To confirm the precision and dependability of the sensors and the machine learning model, data gathered from several test locations was examined.

In over 90% of the cases, the machine learning model's suggestions matched professional agricultural advise, demonstrating the system's great accuracy in forecasting the best crops and fertilizer needs. The smooth tracking of soil health made possible by the cloud API. Real-time monitoring and data visualization features allowed farmers to act quickly and decisively. Furthermore, the system's robustness and flexibility were demonstrated by testing its scalability and low power consumption under various climatic circumstances. Overall, the findings demonstrate how the system can increase agricultural output while

lowering resource waste and encouraging environmentally friendly farming methods.

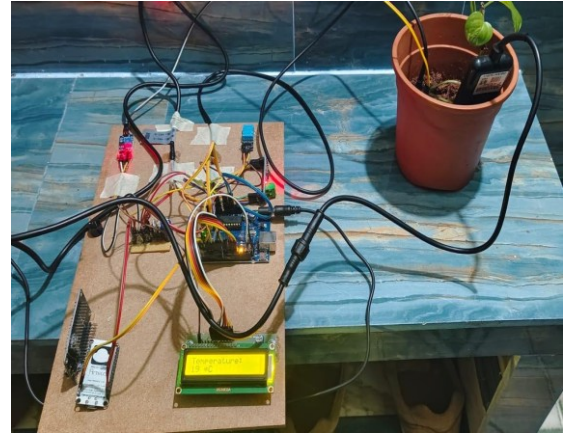


Fig. 11. Full system picture while monitoring soil conditions



Fig. 12. Nitrogen reading on LCD display



Fig. 13. Phosphorus reading on LCD display



Fig. 14. Potassium reading on LCD display



Fig. 15. Soil temperature reading on LCD display

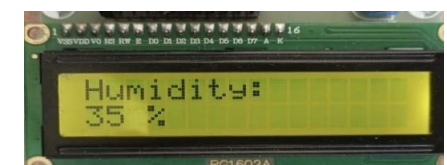


Fig. 16. Humidity reading on LCD display

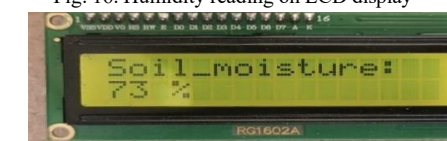


Fig. 17. soil moisture reading on LCD display

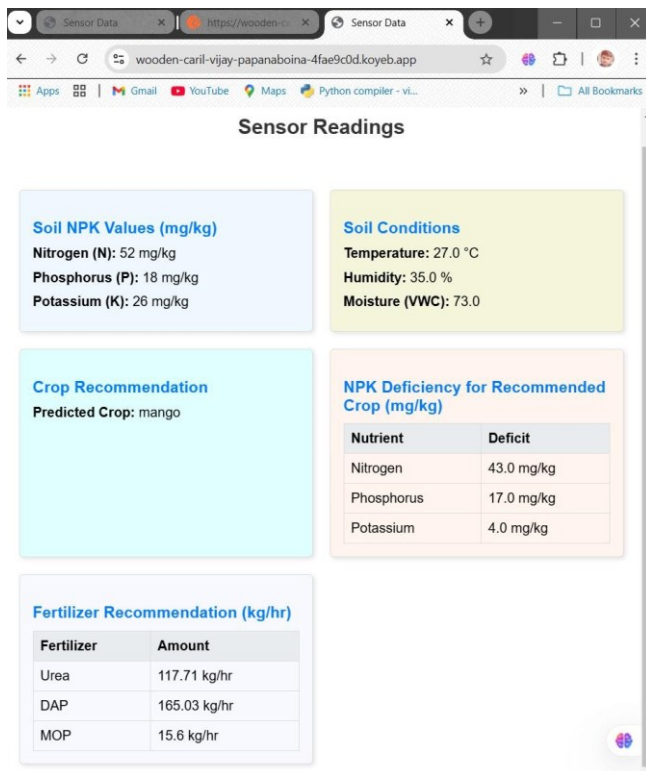


Fig. 18. Readings and predicted data displayed on website

B. Performance Comparison with Traditional Methods

Table 1 : System Characteristics Comparison

Feature	Traditional Soil Testing	Proposed IoT-Based System
Data Collection	It is manually done and time consuming.	It is automatically done in real-time.
Response Time	Delayed (might take days/weeks).	Instantly provide you results.
Farmer Accessibility	Lab analysis is required.	Directly shown in mobile/laptop display.
Decision-Making Support	Relies on farmer's intuition.	AI-based recommendations for precision agriculture.
Cost	High cost due to many times visiting labs for testing.	Low cost due to one time sensor setup.

V. CONCLUSION AND FUTURE SCOPE

An IoT-enabled smart soil monitoring system was presented in this study, integrating real-time data collection, machine learning-based crop and fertilizer recommendations, and data visualization to enhance precision agriculture. The system employs multiple sensors (NPK, soil moisture, temperature, and humidity) interfaced with an Arduino UNO and ESP8266 microcontroller, transmitting data to the cloud for continuous monitoring. The proposed method provides precise, reliable, and actionable insights to farmers, enabling optimal resource utilization, increased crop yield, and sustainable agricultural practices. Experimental results demonstrated that the machine learning model achieved over 90% accuracy in crop and fertilizer recommendations, significantly improving traditional decision-making. Additionally, the system is cost-effective, scalable, and adaptable, addressing key challenges in soil health monitoring, data-driven farming, and environmental sustainability. Future enhancements may include integrating weather prediction models, expanding sensor capabilities to detect heavy metals and micronutrients, and refining machine learning algorithms to further improve accuracy and efficiency. By leveraging IoT and AI-driven analytics, this system contributes to the advancement of precision farming, reducing fertilizer waste, optimizing irrigation, and enhancing soil fertility management, ultimately leading to a more sustainable and productive agricultural ecosystem.

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